

Care Companion

Assistive Companion Mobile Application for Elderly & Disabled Users

Application Documentation & Testing Guide

Platform	Android (Kotlin + Jetpack Compose, minSdk 24)
Backend	Firebase Phone Auth + Supabase (Postgres, RLS, Storage, Realtime, Edge Functions) + FCM
Build	app-debug.apk (27 MB)
Status	Feature-complete; adversarially reviewed and hardened

1. Overview

Care Companion is a dual-role assistive care app. A **Guardian** (family member/caregiver) configures and monitors care for one or more **Elders** from their own phone; each Elder gets a large-type, minimal-tap experience for taking medicines, calling family, recording vitals, and raising an emergency SOS. Everything the guardian saves flows through the cloud to the linked elder's device in real time, and the elder's responses (doses taken, SOS, vitals) flow back to the guardian.

This document maps every module in the project plan to what is implemented in the delivered APK, summarises the architecture, and provides a step-by-step testing guide.

Technology delivered

Concern	Implementation
App	Kotlin + Jetpack Compose, Hilt DI, Navigation Compose, Retrofit
Authentication	Firebase Phone OTP, bridged to Supabase third-party auth (RLS keyed on Firebase UID)
Database	Supabase Postgres — 15 tables, Row-Level Security on every table, RPCs
Images	Supabase Storage (public buckets) + Coil; on-device downsampling
Push	Firebase Cloud Messaging via a Supabase Edge Function (service-account OAuth, FCM v1)
Reminders	On-device exact AlarmManager alarms, reboot-persistent, high-priority notifications
Offline	Room-free local cache + outbox: elder daily loop works without network
Location/SOS	FusedLocationProvider (GPS) + SmsManager (SMS) + dial intents
Vitals report	On-device PDF (PdfDocument) shared via FileProvider
Localisation	Elder UI in English, Hindi, Marathi, Gujarati

Running cost: **Rs. 0/month** on free tiers. The only paid trigger is real OTP SMS beyond the free allowance (test numbers are free and unlimited). FCM push, database, storage and functions are all free at this scale.

2. Feature Inventory (mapped to the plan)

Every functional module from the project proposal, report and expected-screens list is implemented. “Yes” = built & wired to the live backend.

2.1 Authentication & account

Feature	What it does	
Phone OTP login	Sign in with a mobile number and SMS OTP (Firebase). No passwords.	Yes
Role selection	Choose Guardian or Elder at first login; persisted for future sessions.	Yes
Session persistence	Stays logged in across restarts; offline-tolerant cached session.	Yes
Elder verification	Guardian verifies the elder's phone by OTP during profile setup (secondary auth).	Yes
Auto-linking	Elder logs in with the same number and is auto-linked to their profile — no codes.	Yes
Permissions onboarding	One-time explainer + grant for notifications, and location/SMS for elders.	Yes

2.2 Guardian — care management

Feature	What it does	
Dashboard	Multi-elder selector, 10 quick-action tiles, live-updating alert feed.	Yes
Elder profiles	Create, edit, photo, and soft deactivate/reactivate an elder profile.	Yes
Family members	Invite other guardians by phone; Owner / Can-Edit / View-Only access levels.	Yes
Contacts	Add / edit / delete, import from device, mark emergency, photos.	Yes
Medicines	Add / edit / delete, active toggle, dosage, form, with water/milk, meal timing, 3 photos.	Yes
Schedule builder	Day-wise Mon–Sun slots, meal timing, Material time picker.	Yes
Reminders engine	Water / Walk / Vitals + custom categories, each with times and toggle.	Yes
Vitals dashboard	BP / sugar / temp / pulse with Normal/Borderline/High badges, history, PDF export.	Yes
Adherence tracking	Weekly adherence %, per-medicine taken/missed, day strip.	Yes
SOS monitor + settings	Live SOS feed with resolve; choose recipients, edit SMS template, history.	Yes
OTT shortcuts	Configure the video/app tiles shown on the elder's home.	Yes
Wheelchair assistance	Accessible services/places with call and maps shortcuts.	Yes
Guardian settings	Account, notification preference, help, logout.	Yes

2.3 Elder — daily experience (large-type, minimal taps)

Feature	What it does	
Home	Giant SOS button, big labelled tiles, “due today” medicine badge.	Yes

Medicine step-through	One medicine at a time with photo; Taken / Not taken logs adherence.	Yes
Offline dose logging	A Taken/Not-taken tap is queued locally if offline and synced later.	Yes
Contacts	Photo tiles with one-tap dialling.	Yes
Vitals entry / history	Big number fields; readings shown with colour-coded severity.	Yes
Videos (OTT)	Tiles launch the configured apps/links; friendly message if not installed.	Yes
Emergency SOS	5-second countdown → GPS + SMS to contacts + logged; then a big Call button.	Yes
Settings	Text size, high contrast, and language.	Yes
Multi-language	Entire elder experience in English / Hindi / Marathi / Gujarati.	Yes

2.4 On-device & backend automation

Feature	What it does	
Medicine reminders	Exact on-device alarms fire notifications at each dose time; survive reboot.	Yes
FCM push	SOS and missed-dose alerts push to guardians even when the app is closed.	Yes
Missed-dose detection	Server job (every 15 min) marks overdue doses missed and alerts guardians.	Yes
Corrections	If the elder responds late, guardians get a visible correction (never silent).	Yes
SOS fan-out	An SOS instantly fans an alert to every linked guardian.	Yes
Row-Level Security	Guardians see only their linked elders; elders only their own data. 37 tests pass.	Yes

3. How it fits together

The app talks to Supabase's PostgREST API over HTTPS, attaching the signed-in user's **Firestore ID token** to every request. Supabase is configured to trust Firebase as a JWT issuer, so its Row-Level Security policies enforce access using the Firebase user id — a guardian can only read/write elders they are linked to, and an elder only their own record. Privileged operations (creating an elder, verifying a phone, changing ownership) run through SECURITY DEFINER database functions that check ownership. Emergency and missed-dose events are fanned out to per-guardian alert rows; a database trigger calls an Edge Function that sends the FCM push. The elder device arms local alarms for the day and keeps an outbox so the core daily loop works with no connection.

4. Testing Guide

4.1 Prerequisites

- Install **app-debug.apk** on a real Android phone (the app needs Google Play Services for OTP, which emulators without Play Services lack).
- For two-sided testing, use two phones (one Guardian, one Elder) — or one phone, logging out and back in.
- Use the free Firebase **test numbers** (no real SMS): Guardian **+91 98765 00001** code **111111**; Elder **+91 98765 00002** code **222222**; spare **+91 98765 00003** code **333333**.
- Grant notification, location and SMS permissions when the onboarding screen asks.

4.2 Test cases

T1 — Login & roles

- 1 Open the app → pick **Guardian User** → enter +91 98765 00001 → Get OTP → enter 111111 → Login.
- 2 Expected: onboarding screen appears once, then the Guardian dashboard.
- 3 Log out (top-right) and log back in → you land straight on the dashboard (session persists).

T2 — Create & verify an elder

- 1 Dashboard → **Add** (elder selector) → enter name, age, address.
- 2 Enter the elder's phone (+91 98765 00002) → **Send OTP to verify** → enter 222222 → shows "Verified".
- 3 Save. Expected: the new elder appears in the selector; status card shows their name.

T3 — Medicines & schedule

- 1 Select the elder → **Medicines** → + → add "Metformin", 500mg, Tablet, Water, After meal → Save.
- 2 On the medicine card tap **Schedule** → pick days (e.g. every day), meal = Breakfast, tap **Pick time** → 08:00 → Save.
- 3 Tap **Edit** on the card → change the dosage → Save. Expected: the change persists after leaving and returning.

T4 — Contacts

- 1 Select the elder → **Contacts** → + → add a contact (or **Import from Contacts**), mark Emergency → Save.
- 2 Tap the **edit** (pencil) icon → change the name → Save. Expected: updated in the list.

T5 — Elder side on real data

- 1 On the elder device, log in with +91 98765 00002 / 222222 (Elder role).
- 2 Expected: the elder home shows the elder's name, a "due today" badge, and the contacts the guardian added.

- 3 Open **Medicines** → Start taking → mark Taken / Not taken. Expected: guardian's **Adherence** screen reflects it.

T6 — Emergency SOS

- 1 On the elder device tap the big red **SOS** → a 5-second countdown with a large **CANCEL** begins.
- 2 Let it complete. Expected: an SMS goes to the emergency contact(s), the “Alert Sent” screen shows how many were notified, and a big **Call** button appears.
- 3 On the guardian device: the SOS appears in the alert feed / SOS screen (and a push notification if the app is closed).
- 4 Tap SOS again immediately. Expected: a fresh countdown with CANCEL (not a stale “sent” screen).

T7 — Reminders, vitals & PDF

- 1 Guardian → **Reminders** → add a Water reminder with a time → toggle it on.
- 2 Guardian → **Vitals** → add a BP reading (e.g. 150/95) → Expected: it shows with a **High** badge. Tap **Export PDF** → a share sheet opens with the vitals report.
- 3 Elder → **Vitals** → add a reading → Expected: appears in both elder and guardian history.

T8 — Family access & multi-guardian

- 1 Guardian (owner) → **Family** → invite +91 98765 00003 as “View Only” → Send Invite.
- 2 Log in on a third session as +91 98765 00003 (Guardian). Expected: the elder appears; care data is read-only.

T9 — Language & accessibility (elder)

- 1 Elder → **Settings** (gear) → change Language to Hindi → Expected: the whole elder UI switches language.
- 2 Change Text size to Large and toggle High contrast → Expected: text scales up and contrast increases.

T10 — Offline resilience (elder)

- 1 On the elder device, enable Airplane mode → open Medicines → mark a dose Taken.
- 2 Expected: the tap is accepted (no error). Turn network back on and reopen the app → the dose syncs to the guardian.

T11 — Reminders fire on device

- 1 Set a medicine schedule a few minutes in the future → keep the elder app opened once so alarms arm.
- 2 Expected: at the scheduled time a high-priority notification appears with the medicine name.

4.3 Known limitations (non-blocking)

- Real OTP to **non-test** numbers requires the Firebase Blaze plan (free tier of verifications, then per-SMS). Test numbers are free.
- In-app live updates use polling + FCM push; a websocket subscription (sub-second) is a future optimisation.
- The missed-dose scan assumes India (IST); multi-timezone support is future scope.
- The APK is debug-signed — fine for testing/demo; a release keystore is needed for Play Store distribution.